

# Random Signal Theory–EEE 554

## Lecture Notes 3

### Transformations of Two Jointly Distributed Random Variables

**[Reading: Kay, Sections 7.6, 12.6]**

## Transformations of two random variables (RVs)

Type I transformation:

- ▶ Map two RVs to one RV,  $Z = g(X, Y)$

Type II transformation:

- ▶ Map two RVs to two RVs,

$$W = g(X, Y) \text{ and } Z = h(X, Y)$$

## Type I transformation

Given joint PDF  $f_{XY}(x, y)$  (or CDF  $F_{XY}(x, y)$ ) and sample space  $S_{X,Y}$  of  $X$  and  $Y$  find PDF  $f_Z(z)$  or CDF  $F_Z(z)$  of the random variable  $\boxed{Z = g(X, Y)}$

Using the joint PDF of  $X$  and  $Y$ , the CDF of  $Z$  is given by

$$\begin{aligned} F_Z(z) = \Pr(Z \leq z) &= \Pr(g(X, Y) \leq z) \\ &= \Pr((x, y) \in R_z) = \iint_{(x,y) \in R_z} f_{XY}(x, y) \, dx \, dy \end{aligned}$$

where  $R_z$  is a region in  $(x, y)$  plane that contains all points such that  $g(x, y) \leq z$

If  $X$  and  $Y$  are independent, then  $f_{XY}(x, y) = f_X(x)f_Y(y)$  and

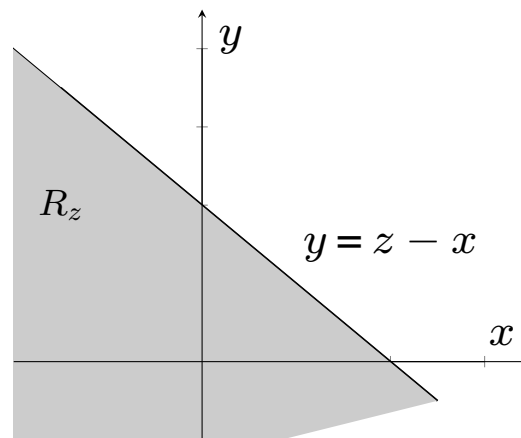
$$F_Z(z) = \iint_{(x,y) \in R_z} f_X(x) f_Y(y) \, dx \, dy$$

Example of simple linear transformation:  $Z = g(X, Y) = X + Y$  (pages 3-6)

To find PDF of  $Z$ , first find the CDF and then take its derivative

$$\begin{aligned} F_Z(z) &= \Pr(Z \leq z) = \Pr(X + Y \leq z) \\ &= \Pr(Y \leq z - x) = \int \int_{(x,y) \in R_z} f_{XY}(x, y) \, dx \, dy \end{aligned}$$

where region  $R_z = \{(x, y) \in S_{XY} : y \leq (z - x), z \in S_Z\}$  (shaded)



To find integration limits in  $R_z$  for

$$F_Z(z) = \int \int_{R_z} f_{XY}(x, y) \, dy \, dx$$

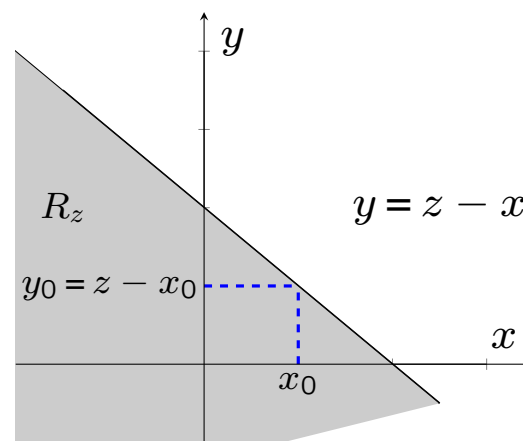
select points  $(x, y) \in S_{X,Y}$  such that  $y \leq (z - x)$  for  $z \in S_Z$

Outer integral (in terms of  $x$ ):  $x = -\infty$  to  $\infty$

Inner integral (in terms of  $y$ ): fix  $x = x_0$ , find range of  $y$  in shaded region

$\Rightarrow$  limits of  $y$  are from  $-\infty$  to  $y_0 = z - x_0$

$$F_Z(z) = \int_{x=-\infty}^{\infty} \left[ \int_{y=-\infty}^{z-x} f_{XY}(x, y) \, dy \right] dx$$



To find PDF of  $Z$ , take derivative of CDF

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz} \left( \int_{x=-\infty}^{\infty} \left[ \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy \right] dx \right)$$

Using the relation  $\frac{d}{dz} \left( \int_{-\infty}^z h(\alpha) d\alpha \right) = h(z)$ , for an arbitrary function  $h(z)$ , then,

$$\frac{d}{dz} \left( \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy \right) = f_{XY}(x, z-x)$$

and the PDF is obtained as 
$$f_Z(z) = \int_{x=-\infty}^{\infty} f_{XY}(x, z-x) dx$$

Important result: for  $X, Y$  independent,  $f_{XY}(x, y) = f_X(x) f_Y(y)$  and

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = f_X(z) \star f_Y(z)$$

where  $\star$  denotes convolution

$\Rightarrow$  for independent RVs, PDF of their sum is the convolution of their PDFs

Continuing example to find CDF of  $Z$  when  $X, Y$  are independent

$$\begin{aligned} F_Z(z) &= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{XY}(x, y) \, dy \, dx \\ &= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_X(x) f_Y(y) \, dy \, dx = \int_{x=-\infty}^{\infty} f_X(x) \left( \int_{y=-\infty}^{z-x} f_Y(y) \, dy \right) dx \end{aligned}$$

Since, by definition, the CDF of  $Y$  is  $F_Y(y) = \int_{-\infty}^y f_Y(\zeta) \, d\zeta$ ,

$$\begin{aligned} F_Z(z) &= \int_{x=-\infty}^{\infty} f_X(x) \left( \int_{y=-\infty}^{z-x} f_Y(y) \, dy \right) dx \\ &= \boxed{\int_{x=-\infty}^{\infty} f_X(x) F_Y(z - x) \, dx} \end{aligned}$$